

JEE-Main-23-01-2025 (Memory Based)

[MORNING SHIFT]

Physics

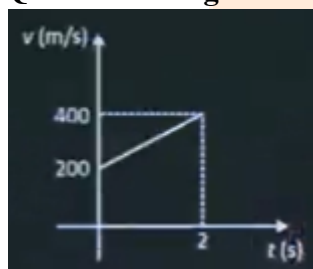
Question: If flux can be represented as $\phi = \lambda\alpha + \beta\sigma$ where λ is the linear charge density and σ is the surface charge density then the dimensions of α/β is that of

Options:

- (a) Displacement
- (b) 1/Displacement
- (c) 1/Area
- (d) Area

Answer: (b)

Question: For given velocity-time (v-t) graph, find distance travelled at 0.5 sec



Options:

- (a) 125 m
- (b) 112.5 m
- (c) 137.5 m
- (d) 150 m

Answer: (b)

Question: The displacement of a particle as function of time is $x(t) = A \sin(t) + B \cos^2(t)$

+ $ct^2 + D$. Find dimension of $\left[\frac{ABC}{D} \right]$

Options:

- (a) L^2
- (b) L^2T^{-2}
- (c) LT^{-2}
- (d) L^3T

Answer: (b)

Question: Two points represented by position vectors

$2\hat{i} + 3n\hat{j} - 2\hat{k}$ and $2\hat{i} + 4p\hat{j} - 2\hat{k}$ Same distance from origin. If the vectors are perpendicular then find n.

Options:

- (a) $\pm \frac{2\sqrt{2}}{3}$
- (b) $\pm \frac{3}{2\sqrt{2}}$
- (c) $\pm \frac{2}{3}$
- (d) $\pm \frac{1}{\sqrt{2}}$

Answer: (a)

Question: Match the column (1) and (2)

column (1)	column (2)
(a) Temperature remains constant in a process	(i) Adiabatic process
(b) Pressure remains constant in a process	(ii) Isobaric process
(c) Volume remains constant in a process	(iii) Isothermal process
(d) There is no heat exchange in the process	(iv) Isochoric process

Options:

- (a) (a)- (iii) (b) -(ii) (c) - (iv) (d) -(i)
- (b) (a)- (ii) (b) -(iii) (c) - (i) (d) -(iv)
- (c) (a)- (i) (b) -(iii) (c) - (ii) (d) -(iv)
- (d) (a)- (iv) (b) -(iii) (c) - (i) (d) -(ii)

Answer: (a)

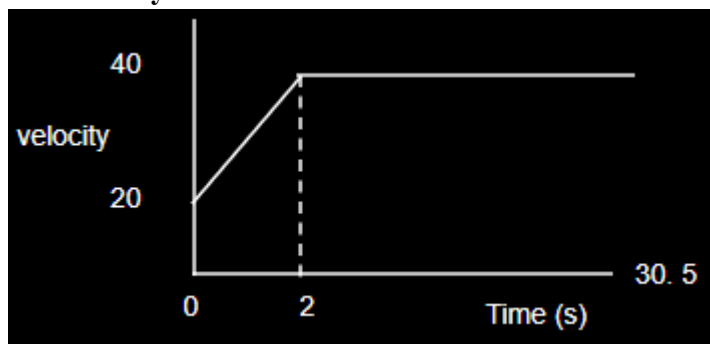
Question: A lead bullet of mass 0.01g moving with a speed of 100m/s enters a fixed wooden block. If no heat is taken away by the wood, the rise in temperature of the bullet in the wood is nearly (Sp. Heat of lead = 150cal/kg, Melting point of lead = 600K)

Options:

- (a) 8 K
- (b) 10 K
- (c) 4 K
- (d) 5 K

Answer: (a)

Question: The velocity time graph of a moving car is given below, find the total distance travelled by the car in 30.5 seconds



Options:

- (a) 5900 m
- (b) 12000 m
- (c) 14000 m
- (d) 16000 m

Answer: (b)

Question: An ideal gas is suddenly compressed to 1/4th of its original volume. Ratio of specific heat at constant pressure and constant volume is 3/2. What is the rise in the temperature of gas, the original temperature being 0 degree celsius.

Options:

- (a) 546 K
- (b) 273° C
- (c) 546° C
- (d) 600 K

Answer: (b)

Question: The ratio of electric force to gravitational force between two particles having charges q_1 q_2 and masses m_1 and m_2 respectively is (where symbols have their usual meanings)

Options:

- (a) $\frac{4\pi\epsilon_0 m_1 m_2 G}{q_1 q_2}$
- (b) $\frac{4\pi\epsilon_0 G m_1 m_2}{q_1 q_2 r^4}$
- (c) $\frac{4\pi\epsilon_0 G m_1 m_2}{q_1 q_2}$
- (d) $4\pi\epsilon_0 G m_1 m_2$

Answer: (d)

Question: Solid sphere of mass m rolls down from rest and achieves speed V_1 on an inclined plane of 30° sphere achieves speed V_2 in case of pure rolling on a inclined plane of 45° when released from same height. Then the ratio V_1/V_2 is (Assume pure rolling)

Options:

- (a) 1
- (b) $\frac{2}{5}$
- (c) $\frac{3}{5}$
- (d) $\frac{7}{5}$

Answer: (a)

Question: In a moving-coil galvanometer, if the number of turns are increased then.

- 1) Current Sensitivity increases
- 2) Resistance of galvanometer increases
- 3) Voltage sensitivity decrease
- 4) A Galvanometer can be connected to an ammeter by connecting a very resistance in parallel with it.

Select option having only correct statement from the above list

Options:

- (a) 1, 2, 3
- (b) 3, 4
- (c) 1, 2, 4
- (d) 1, 2, 3, 4

Answer: (c)

Question: Statement-1 : Hot water flows faster than cold water

Statement -2: Surface tension of soap bubble is more than that of a drop of water

Options:

- (a) Both Statements are correct
- (b) Statement 1 is true and 2 is false
- (c) Statement 1 is false and 2 is true
- (d) Both Statements are false

Answer: (b)

Question: A circular disc of diameter 10 cm is cut from one edge of a larger circular disc of radius 20 cm. The shift in centre of mass is

Options:

- (a) 1 cm
- (b) -1 cm
- (c) 2 cm
- (d) -2 cm

Answer: (b)

Question: (1) Current $I_2 = 1$ A

(2) Current $I_1 = I_4 = 2$ A

(3) Current $I_3 = 0.5$ A

(4) Current $I_4 = 1$ A

(5) Diode is forward bias condition

Study the above circuit diagram and select only the correct statement/s from the above list.

← 1/3 Gray →

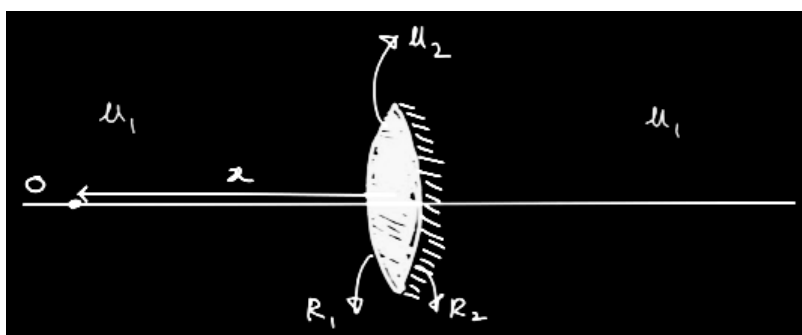
(a) $4\pi \times 10^{-2} \text{ s}$

- (b) $2\pi \times 10^{-2}$ s
 (c) $\pi \times 10^8$ s
 (d) $2\pi \times 10^{-3}$ s

Answer: (b)

Question:

A thin biconvex lens having two surfaces of radii of curvature R_1 and R_2 is made of a material of refractive index μ_c . It is kept in a medium of refractive index μ_1 . Now the outer surface lens having radius R_2 is silvered. If a point object is placed on principal axis at a distance of x from lens and its final image is coinciding with object itself, then find x .



Options:

- (a) $x = \frac{2\mu_1 R_1 R_2}{2R_2(\mu_2 + \mu_1) + R_1\mu_2}$
 (b) $x = \frac{2\mu_1 R_1 R_2}{2R_2(\mu_2 - \mu_1) + R_1\mu_2}$
 (c) $x = \frac{2\mu_1 R_1 R_2}{2R_2(\mu_2 + \mu_1) + R_1\mu_2}$
 (d) $x = \frac{2\mu_1 R_1 R_2}{R_2(\mu_2 + \mu_1) - R_1\mu_2}$

Answer: (b)

Question: If the $\vec{E}$ in an EM wave is given by $\vec{E} = (4\hat{i} - 3\hat{j})$

$\cos(\omega t - (3x + 4y)k)$ then the magnetic field $\vec{B}$ will be

Options:

- (a) $\vec{B} = -\left(\frac{5}{3} \times 10^{-8}\right)\hat{k} \cos(\omega t - (3x + 4y)k)$
 (b) $\vec{B} = \left(\frac{5}{3} \times 10^{-8}\right)\hat{k} \cos(\omega t - (3x + 4y)k)$
 (c) $\vec{B} = -\frac{5}{3} \times 10^{-8}\hat{j} \cos(\omega t - (3x + 4y)k)$

(d) $\vec{B} = \frac{5}{3} \times 10^{-8} \hat{j} \cos(\omega t - (3x + 4y)k)$

Answer: (a)



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[MORNING SHIFT]

Chemistry

Question: Which of the following React with Hinsberg reagent?

- (i) Aniline
- (ii) N, N-Dimethylaniline
- (iii) Methyl amine
- (iv) $C_6H_5NHC_6H_5$

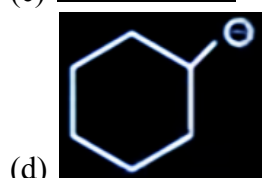
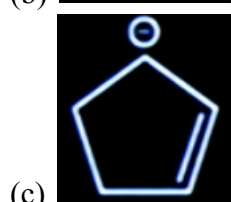
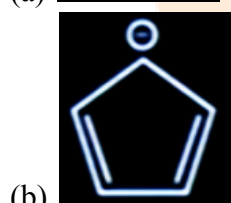
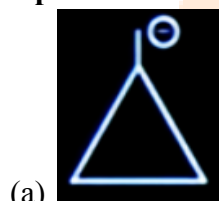
Options:

- (a) i only
- (b) i and iii only
- (c) i, iii and iv
- (d) i and ii only

Answer: (c)

Question: Among the following, the most stable carbanion is

Options:



Answer: (b)

Question: Which of the following compound can show fac-mer isomerism?

Options:

- (a) $[Co(NH_3)_6]Cl_3$
- (b) $[Co(NH_3)_3Cl_3]$
- (c) $[Co(en)_2(NH_3)_2]Cl_2$

(d) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$

Answer: (b)

Question: Which of the following pair of ions are same coloured?

Options:

- (a) Ti^{4+} , V^{3+}
- (b) Cr^{2+} , Cu^{2+}
- (c) Cr^{3+} , Ni^{2+}
- (d) Mn^{3+} , Fe^{2+}

Answer: (b)

Question: Which of the following does not belong to the same period in the modern periodic table?

Options:

- (a) Pd
- (b) Ir
- (c) Pt
- (d) Os

Answer: (a)

Question: Statement-1: Fructose can give tollens test even though it does not have aldehyde group

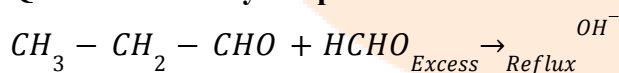
Statement-2: When reacted with base fructose can undergo rearrangement to produce aldehyde group

Options:

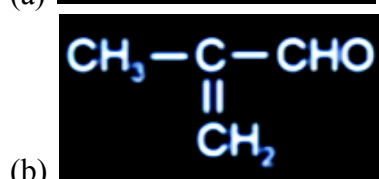
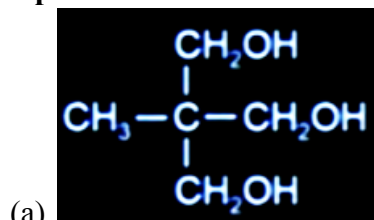
- (a) If both Statement-1 and Statement-2 are true and the Statement-2 is the correct explanation of the assertion
- (b) If both Statement-1 and Statement-2 are true but Statement-2 is not the correct explanation of the assertion
- (c) If Statement-1 is true but Statement-2 is false
- (d) If the statement-1 and Statement-2 both are false

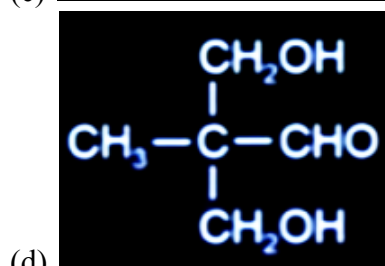
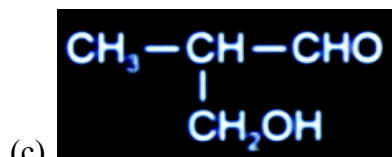
Answer: (b)

Question: Identify the product formed in the following reaction



Options:





Answer: (a)

Question: Incorrect statement among the following is

Options:

- (a) SO_2 act as oxidizing agent but not reducing agent
- (b) NO_2 exists as dimer
- (c) PF_3 exists but NF_5 does not
- (d) PH_3 has lower proton affinity than NH_3

Answer: (a)

Question: Match the column appropriately regarding thermodynamics process.

	Column-I		Column-II
(P)	When volume change is zero	(i)	$\Delta w = 0$
(Q)	When pressure is constant	(ii)	$\Delta Q = 0$
(R)	When no heat is exchanged	(iii)	Isobaric
(S)	Work done by the gas is equal to heat given to the gas	(iv)	Isothermal

Options:

- (a) P(iv), Q(iii), R(i), S(ii)
- (b) P(i), Q(iii), R(ii), S(iv)
- (c) P(ii), Q(iii), R(iv), S(i)
- (d) P(ii), Q(iii), R(i), S(iv)

Answer: (b)

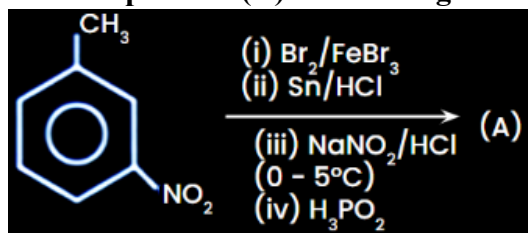
Question: $\text{Co}(\text{NH}_3)_x\text{Cl}_3$ has 0.1 molal. 100% Dissociation $\Delta T_f = 0.558$ ($k_f = 1.86$) Then formula of compound is

Options:

- (a) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
- (b) $[\text{Co}(\text{NH}_3)_2\text{Cl}_4]$
- (c) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- (d) None

Answer: (c)

Question: Consider the following sequence of reactions and find the molecular mass of the final product (A) formed in g mol^{-1} .



Options:

- (a) 108
- (b) 216
- (c) 171
- (d) 186

Answer: (c)

Question: Match the Column-I with Column-II and choose the correct

	Column-I		Column-II
(A)	BF_3	(i)	Odd e- species
(B)	$\text{CCl}_4, \text{CO}_2$	(ii)	Expanded octet
(C)	$\text{PCl}_5, \text{BrF}_5$	(iii)	Complete octet
(D)	NO	(iv)	Electron deficient

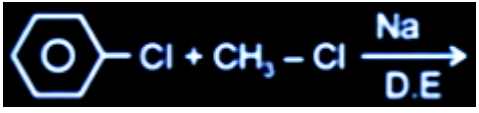
Options:

- (a) A-iii, B-iv, C-i, D-ii
- (b) A-iv, B-ii, C-iii, D-i
- (c) A-iv, B-iii, C-ii, D-i
- (d) A-i, B-ii, C-iii, D-iv

Answer: (c)

Question: Match the column and choose the correct option

	Column-I		Column-II
(A)		(P)	Sandmeyer reaction
(B)		(Q)	Fitting reaction

(C)		(R)	Wurtz-Fittig reaction
(D)	$\text{CH}_3 - \text{Cl} + \text{AgF} \rightarrow$	(S)	Swart's reaction

Options:

- (a) A-Q, B-P, C-R, D-S
- (b) A-Q, B-P, C-S, D-R
- (c) A-Q, B-R, C-S, D-P
- (d) A-Q, B-P C-R, D-S

Answer: (d)

Question: Co^{2+} is forming an octahedral complex with spin only magnetic moment 3.83 BM. Which of the following electronic configurations?

Options:

- (a) $t_{2g}^5 e_g^2$
- (b) $t_{2g}^6 e_g^1$
- (c) $t_{2g}^4 e_g^3$
- (d) $e_g^4 t_{2g}^3$

Answer: (a)

Question: Given below are two statements:

Statement-I: During Lassaigne's test, covalent compound is converted to ionic compound

Statement-II: $\text{Na}_4[\text{Fe}(\text{CN})_6]$ gives Prussian blue colour on reaction with $\text{Fe}_2(\text{SO}_4)_3$

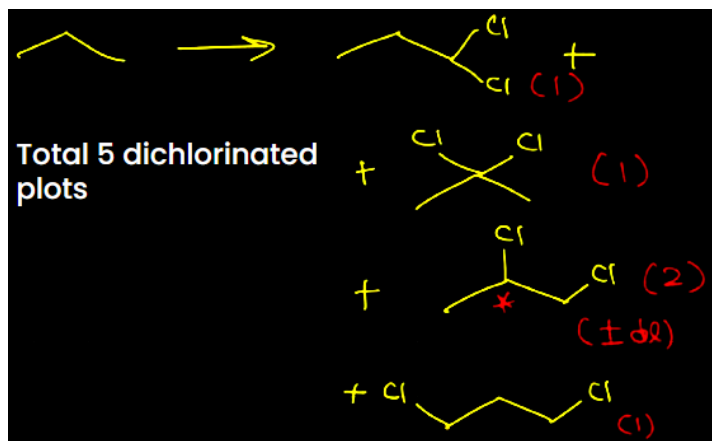
Options:

- (a) S-I is correct, S-II is incorrect
- (b) S-I is incorrect, S-II is correct
- (c) Both S-I and S-II are correct
- (d) Both S-I and S-II are incorrect

Answer: (c)

Question: How many dichlorinated products can be obtained for n-propane?

Answer: (5)



Question: Consider the following $FeO_4^{2-} \xrightarrow{2.0V} Fe^{3+} \xrightarrow{0.8V} Fe^{2+} \xrightarrow{-0.5V} Fe^0$

Find $E^0_{FeO_4^{2-}/Fe^{2+}}$ (Nearest integer)

Answer: (2)

Solution:

$$E^0 = \frac{n_1 E_1 + n_2 E_2}{n_1 + n_2} = \frac{(3)(2) + (1)(0.8)}{4} = \frac{6.8}{4} = 1.7V$$

Nearest integer = 2

Question: Consider the given values:

$$\Delta H = 55 \text{ kJ mol}^{-1}$$

$$\Delta S = 175 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$T = 25^\circ\text{C}$$

Calculate the value of Gibbs free energy charge (ΔG) in J mol^{-1} .

Answer: (2850)

Question: Calculate the percentage by weight of S if 160 gram of organic compound produce 466 gram of $BaSO_4$

Answer: (40%)

Question: If 2gm phenol is allowed to react with Br_2/H_2O . How much Br_2 will be required to produce 2, 4, 6 tribromophenol (Rounded off to nearest integer)

Answer: (10)

Question: If 2gm phenol is allowed to react with Br_2/H_2O . How much Br_2 will be required to produce 2, 4, 6 tribromophenol (Rounded off to nearest integer)

Answer: (179)

Solution:

$$\left(\frac{x+10^{-3}}{44}\right) - \frac{10^{21}}{6.022 \times 10^{23}} = 2.4 \times 10^{-3}$$

$x = 179$



JEE-Main-23-01-2025 (Memory Based)
[MORNING SHIFT]
Math

Question: How many words can be formed from the word DAUGHTER such that any vowels are not together

Options:

- (a) 14444
- (b) 14400
- (c) 10044
- (d) 12400

Answer: (b)

DGHTR AUE

$$\begin{aligned}
 &5! \times {}^6C_3 \times 3! \\
 &= 120 \times 20 \times 6 \\
 &= 14400
 \end{aligned}$$

Question: Two biased dice are tossed. Die 1 has 1 on two faces, 2 on two faces, 3 and 4 on other faces, while die 2 has 2 on 2 faces, 4 on 2 faces and 1 and 3 on other faces. Then the probability that when throwing these dice we get sum 4 or 5.

Options:

- (a) $\frac{3}{7}$
- (b) $\frac{2}{3}$
- (c) $\frac{4}{9}$
- (d) $\frac{8}{9}$

Answer: (c)

1 on 2 2 on 2

2 on 2 4 on 2

3 on 1 1 on 1

4 on 1 3 on 1

Sum (4 or 5)

$$= P(1, 3 \text{ or } 4) + P(2, 2 \text{ or } 3)$$

$$+ P(3, 1 \text{ on } 2) + P(4, 1)$$

$$\begin{aligned}
 &\frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{2} + \frac{1}{6} \times \frac{1}{2} + \frac{1}{6} \times \frac{1}{6} \\
 &= \frac{1}{6} + \frac{1}{6} + \frac{1}{12} + \frac{1}{36} = \frac{16}{36} = \frac{4}{9}
 \end{aligned}$$

Question: If for an arithmetic progression if first term is 3 and sum of first four terms is equal to $\frac{1}{5}$ of the sum of next four terms, then the sum of first 20 terms is

Options:

- (a) 1080
- (b) 364

(c) -1080

(d) -364

Answer: (c)

$$S_4 = \frac{1}{5}(S_8 - S_4)$$

$$5S_4 = S_8 - S_4$$

$$6S_4 = S_8$$

$$6 \times \frac{4}{2}[6 + 3d] = \frac{8}{2}[6 + 7d]$$

$$\Rightarrow 12(6 + 3d) = 4(6 + 7d)$$

$$3(6 + 3d) = (6 + 7d)$$

$$2d = 6 - 18$$

$$d = -6$$

$$\text{Now } S_{20} = \frac{20}{2}[6 + 19(-6)]$$

$$= 10 \times 6(1 - 19)$$

$$= 60 \times -18 = -1080$$

**Question: If $f(x)$ is continuous at $x = 0$, where
Then $k_1^2 + k_2^2$ is**

Options:

(a) 1

(b) 2

(c) 0

(d) 4

Answer: (b)

$$LHL_{x=0} = \lim_{x \rightarrow 0^-} 2 \left[\frac{\sin(k_1+1)x}{x} + \frac{\sin(k_2+1)x}{x} \right]$$

$$= 2[k_1 + 1 + k_2 + 1] = 2(k_1 + k_2 + 2)$$

$$RHL_{x=0} = \lim_{x \rightarrow 0^+} \frac{2}{x} \log \left(\frac{k_2 x + 1}{k_1 x + 1} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{2}{x} \left(\frac{k_2 x + 1}{k_1 x + 1} - 1 \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{2}{x} \left[\frac{x(k_2 - k_1)}{k_1 x + 1} \right]$$

$$RHL_{x=0} = 2(k_2 - k_1)$$

$$f(0) = 4$$

$$\Rightarrow 2(k_2 - k_1) = 4 = 2(k_1 + k_2 + 2)$$

$$k_2 - k_1 = 2 \quad k_2 = 1$$

$$k_1 + k_2 = 0 \quad k_1 = -1$$

$$k_1^2 + k_2^2 = 2$$

$$f(x) = \begin{cases} \frac{2}{x}(\sin(k_1 + 1)x + \sin(k_2 + 1)x) & x < 0 \\ 4 & x = 0 \\ \frac{2}{x} \log \left[\frac{k_2 x + 1}{k_1 x + 1} \right] & x > 0 \end{cases}$$

Question: $(a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ has equal roots $a + c = 15$; $b = \frac{36}{5}$ find $a^2 + c^2 = ?$

Options:

- (a) 108
- (b) 225
- (c) 117
- (d) 333

Answer: (c)

$$a + c = 15 \quad b = \frac{36}{5}$$

$$\frac{c(a-b)}{a(b-c)} = 1 \rightarrow ac - bc = ab - ac$$

Assuming roots are repeated

$$2ac = b(a + c)$$

$$= \frac{36}{5} \times 15$$

$$a^2 + c^2 = 15^2 - 108$$

$$= 117$$

Question: value of $\cos^{-1} \left[\frac{12}{13} \cos x + \frac{5}{13} \sin x \right]$ is $\left(x \in \left[\frac{\pi}{2}, \pi \right] \right)$

Options:

- (a) $x + \tan^{-1} \frac{12}{13}$
- (b) $x - \tan^{-1} \frac{12}{13}$
- (c) $x - \tan^{-1} \frac{5}{12}$
- (d) $x + \tan^{-1} \left(\frac{4}{5} \right)$

Answer: (c)

$$= \cos^{-1} [\cos(x - \alpha)] \quad \tan \alpha = \frac{5}{12} \alpha \in 1^{st} qu$$

$$= x - \alpha$$

$$= x - \tan^{-1} \frac{5}{12}$$

Question: If for the system of linear equations having infinite solutions $(\lambda - 4)x + (\lambda - 2)y + \lambda z = 0$

$$2x + 3y + 5z = 0$$

$$x + 2y + 6z = 0 \text{ then } \lambda^2 + \lambda \text{ is}$$

Options:

- (a) 9
- (b) 81
- (c) 90
- (d) 162

Answer: (c)

$$\begin{vmatrix} \lambda - 4 & \lambda - 2 & \lambda \\ 2 & 3 & 5 \\ 1 & 2 & 6 \end{vmatrix} = 0$$

$$\rightarrow \begin{vmatrix} \lambda - 4 & 6 - \lambda & 6 - 2\lambda \\ 2 & -1 & -4 \\ 1 & 0 & 0 \end{vmatrix} = 0$$

$$4(\lambda - 6) + (6 - 2\lambda) = 0$$

$$2\lambda - 18 = 0$$

$$\lambda = 9$$

$$\lambda^2 + \lambda = 90$$

Question: A relation defined on set $A = \{1, 2, 3, 4\}$, then how many minimum ordered pairs are added to $R = \{(1, 2), (2, 3), (3, 3)\}$ so that it becomes an equivalence relation ?

Options:

- (a) 4
- (b) 5
- (c) 6
- (d) 7

Answer: (d)

The given set is $A = \{1, 2, 3, 4\}$, and the relation $R = \{(1, 2), (2, 3), (3, 3)\}$. To make R an equivalence relation, it must satisfy reflexivity, symmetry, and transitivity.

1. **Reflexivity:** Add $(1, 1), (2, 2), (4, 4)$.
2. **Symmetry:** Add $(2, 1), (3, 2)$ (to match $(1, 2)$ and $(2, 3)$).
3. **Transitivity:**
 - $(1, 2)$ and $(2, 3)$ imply $(1, 3)$.
 - $(2, 3)$ and $(3, 2)$ imply $(2, 2)$ (already added for reflexivity).
 - $(3, 2)$ and $(2, 1)$ imply $(3, 1)$.

Total pairs to add:

$$\{(1, 1), (2, 2), (4, 4), (2, 1), (3, 2), (1, 3), (3, 1)\}.$$

Number of pairs added: 7.

Question: The sum of all rational terms in the expansion of $\left(1 + 2^{\frac{1}{3}} + 3^{\frac{1}{2}}\right)^6$ is

Options:

- (a) 612
- (b) 373
- (c) 400
- (d) 180

Answer: (a)

$$\left(1 + 2^{\frac{1}{3}} + 3^{\frac{1}{2}}\right)^6$$

$$\frac{6!}{a!b!c!} 1^a 2^{\frac{b}{3}} 3^{\frac{c}{2}}; a + b + c = 6$$

$$b \rightarrow 0, 3, 6$$

$$c \rightarrow 0, 2, 4, 6$$

$$a \quad b \quad c$$

$$6 \quad 0 \quad 0$$

$$4 \quad 0 \quad 2$$

$$2 \quad 0 \quad 4$$

$$0 \quad 0 \quad 6$$

$$3 \quad 3 \quad 0$$

$$1 \quad 3 \quad 2$$

$$0 \quad 6 \quad 0$$

$$\frac{6!}{6!} + \frac{6!}{4!2!} 3 + \frac{6!}{2!4!} + \frac{6!}{6!} 3^3 + \frac{6!}{3!3!} 2^1 + \frac{6!}{3!2!} 2!3! + \frac{6!}{6!} 2^2$$

$$= 1 + 45 + 135 + 27 + 40 + 360 + 4$$

$$= 612$$

Question: $(7 + e^{2x})dy - (1 + 2e^{2x})(y + 3)dx = 0$, $y(0) = 5$ then $y(\ln 2)$

Solution :

$$\frac{dy}{y+3} = \frac{1+2e^{2x}}{7+e^{2x}} dx = \frac{e^{-2x}}{7e^{-2x}+1} + \frac{2e^{2x}}{7+e^{2x}} dx$$

$$\ln|y+3| = -\frac{1}{2}\ln|1+7e^{-2x}| + \ln|7+e^{2x}| + C$$

$$\ln 8 = -\frac{1}{2}\ln|8| + \ln 8 + c \quad c = \frac{3}{2}\ln 2$$

$$\ln(y+3) = -\frac{1}{2}\ln\left|1+7\cdot\frac{1}{4}\right| + \ln|7+4| + \frac{3}{2}\ln 2$$

$$= -\frac{1}{2}\ln\frac{11}{4} + \ln 11 + \frac{3}{2}\ln 2$$

$$= \frac{1}{2}\ln\left(\frac{121 \times 4}{11} \times 8\right) = \frac{1}{2}\ln(11 \times 32)$$

$$y+3 = 4\sqrt{22}$$

$$y = 4\sqrt{22} - 3$$

Question: The range of values of a for which $5x^3 - 15x - a = 0$ has 3 distinct solutions is (α, β) then $\beta - 2\alpha$ is

Options:

- (a) 10
- (b) 20
- (c) 30
- (d) 40

Answer: (c)

$$f'(x) = 15x^2 - 15 = 0$$

$$x = \pm 1$$

$$f(1)f(-1) = -(10+a)(10-a) < 0$$

$$a \in (-10, 10)$$

$$\beta - 2\alpha = 30$$

Question: Find the value of $\sin 70^\circ (\cot 10^\circ \cot 70^\circ - 1)$

Options:

(a) 0

(b) 1

(c) 2

(d) 3

Answer: (b)

$$\sin 70^\circ (\cot 10^\circ \cot 70^\circ - 1)$$

$$\cot 80 = \frac{(\cot 10 \cot 70^\circ - 1)}{\cot 10 + \cot 70}$$

$$= \sin 70^\circ \left[\frac{1}{\sqrt{3}} (\cot 10 + \cot 70) \right] \cot 80$$

$$= \sin 70 \left[\frac{\sin 80}{\sin 10 \sin 70} \right] \frac{\cos 80}{\sin 80}$$

$$= 1$$

Question: Area of the larger region bounded by curves $y = |x - 1|$ and $x^2 + y^2 = 25$ is

Solution :

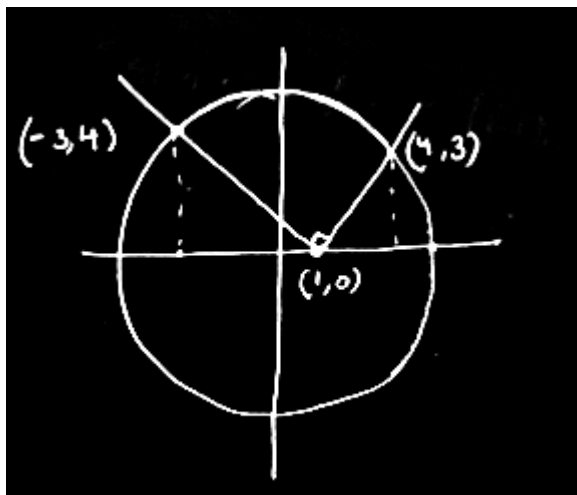
$$A = 25\pi - \int_3^1 \sqrt{25 - x^2} - (1 - x) dx$$

$$- \int_{-3}^1 \sqrt{25 - x^2} - (x - 1) dx$$

$$= 25\pi - \int_3^4 \sqrt{25 - x^2} dx + 8 + \frac{9}{2}$$

$$= 25\pi - \left[\frac{x}{2} \sqrt{25 - x^2} + \frac{25}{2} \sin^{-1} \frac{x}{5} \right]_{-3}^4 + \frac{25}{2}$$

$$- 25\pi - \left[(6 + 3) + \frac{25}{2} \left(\frac{\pi}{2} \right) \right] + \frac{25}{2} = \frac{75\pi}{4} + \frac{7}{2}$$



Question: $f(x) = \log_e x$ $g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$ Then find the domain of $\log(x)$.
Options:

- (a) Q
- (b) R
- (c) C
- (d) N

Answer: (b)

$$\frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1} > 0$$

$$x^4 - 2x^3 + 3x^2 - 2x + 2 > 0$$

$$\rightarrow (x^2 + 1)(x^2 + 2) - 2x(x^2 + 1) > 0$$

$$(x^2 + 1)[x^2 - 2x + 2] > 0$$

So $x \in R$.

$$f(x) = \int \frac{dx}{(x-1)^{\frac{11}{13}}(x+15)^{\frac{15}{13}}} \text{ and } I(37) - I(24) = \frac{1}{4} \left(\frac{1}{b^{\frac{1}{13}}} - \frac{1}{c^{\frac{1}{13}}} \right) \text{ the value of}$$

Question:
3(b + c)

Options:

- (a) 9
- (b) 4
- (c) 27
- (d) 39

Answer: (d)

$$\int \frac{dx}{(x-1)^2 \left[\left(\frac{x+15}{x-11} \right)^{\frac{15}{3}} \right]}$$

$$\frac{x+15}{x-11} = t$$

$$\Rightarrow \frac{(x-11)1 - (x+15)1}{(x-1)^2}$$

$$\frac{dx}{(x-1)^2} = \frac{-1}{26}$$

$$\Rightarrow \frac{-1}{26} \int t^{\frac{-15}{13}} dt = \frac{-1}{26} \cdot \frac{t^{\frac{-2}{13}}}{\frac{-2}{13}}$$

$$I = \frac{1}{4} \left(\frac{x-11}{x+15} \right)^{\frac{2}{13}} + C$$

$$I(37) = \frac{1}{4} \left[\frac{26}{52} \right]^{\frac{2}{13}} = \frac{1}{4} 2^{\frac{2}{13}} = \frac{1}{4} \cdot 4^{\frac{1}{13}}$$

$$I(24) = \frac{1}{4} \left[\frac{13}{39} \right]^{\frac{2}{13}} = \frac{1}{4} \cdot 3^{\frac{2}{13}} = \frac{1}{4} \cdot 9^{\frac{1}{13}}$$

$$b = 4_1 C = 9$$

$$3(b + c) = 39$$

Question: Two vertices of a triangle are $(-2, 4)$ and $(5, 4)$ and its orthocenter is $(3, 5)$. The area of triangle is 35 and centroid is (c, d) then $c + 2d$ is equal to

Options:

(a) $10/3$

(b) $20/3$

(c) $40/3$

(d) $50/3$

Answer: (d)

Now, Area of $\triangle ABC$

$$= \frac{1}{2} AB \times CP = 3.5$$

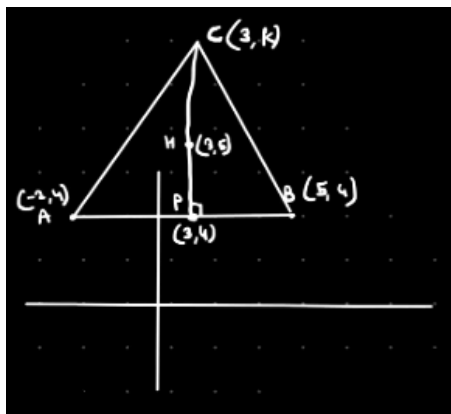
$$\frac{1}{2} \times 7 \times (k - 4) = 35$$

$$k = 10$$

$$\text{Now centroid} \left(\frac{3+5-2}{3}, \frac{4+4+14}{3} \right)$$

$$\left(2, \frac{22}{3} \right)$$

$$c + 2d = 2 + \frac{44}{3} = \frac{50}{3}$$



Question:
$$\int_{e^2}^{e^4} \frac{e^{(\ln^2 x + 1)^{-1}}}{e^{(\ln^2 x + 1)^{-1}} + e^{((6 - \ln x)^2 + 1)^{-1}}} \frac{dx}{x}$$

Options:

- (a) 4
- (b) 2
- (c) 0
- (d) 1

Answer: (d)

$$\int_{e^2}^{e^4} \frac{e^{(\ln^2 x + 1)^{-1}}}{e^{(\ln^2 x + 1)^{-1}} + e^{((6 - \ln x)^2 + 1)^{-1}}} \frac{dx}{x}$$

$$\ln x = t \quad \frac{4}{2}$$

$$\frac{1}{n} dn = dt$$

$$\int_2^4 \frac{e^{(t^2 + 1)^{-1}}}{e^{(t^2 + 1)^{-1}} + e^{((6 - t)^2 + 1)^{-1}}} dt$$

$$t \rightarrow 6 - t$$

$$2I = \int_2^4 1 dt$$

$$2I = (4 - 2)$$

$$I = 1$$

Question: Let Q be the foot of perpendicular from point P(3,4,5) the line

$$\frac{x-1}{2} = \frac{y+3}{-5} = \frac{z-5}{2}. \text{ Find the area of } \Delta PQR \text{ given that } R(1, -2, 3).$$

Solution :

$$L : \frac{x-1}{2} = \frac{y+3}{-5} = \frac{z-5}{2} = \lambda$$

$$(2\lambda + 1, -5\lambda - 3, 2\lambda + 5) \quad \vec{b} = 2\hat{i} - 5\hat{j} + 2\hat{k}$$

Now

$$\overrightarrow{PQ} = (2\lambda - 2)\hat{i} + (-5\lambda - 7)\hat{j} + 2\lambda\hat{k}$$

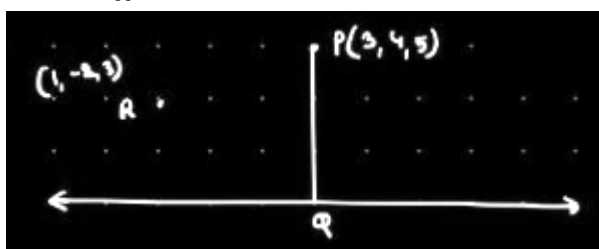
$$\text{Now } \overrightarrow{PQ} \perp \vec{b}$$

$$\Rightarrow \overrightarrow{PQ} \cdot \vec{b} = 0$$

$$2(2\lambda - 2) - 5(-5\lambda - 7) + 4\lambda = 0$$

$$33\lambda + 31 = 0$$

$$\Rightarrow \lambda = -\frac{31}{33}$$



$$Q\left(-\frac{29}{33}, -\frac{204}{39}, \frac{103}{33}\right)$$

$$\text{Now } \overrightarrow{RP} = 2\hat{i} + 6\hat{j} + 2\hat{k}$$

$$\overrightarrow{RQ} = -\frac{62}{33}\hat{i} - \frac{138}{33}\hat{j} + \frac{4}{33}\hat{k}$$

$$\text{Now area of } \triangle PQR = \frac{1}{2} \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 6 & 2 \\ -\frac{62}{33} & -\frac{138}{33} & \frac{4}{3} \end{vmatrix} \right\|$$

$$= \frac{1}{33} \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 1 \\ -62 & -138 & 4 \end{vmatrix} \right\|$$

$$= \frac{1}{33} |150\hat{i} - 66\hat{j} - 48\hat{k}|$$

$$= \frac{1}{33} \sqrt{22500 + 4356 + 2304}$$

$$= \frac{1}{3} \sqrt{29160}$$



$$\left| \frac{z}{z+i} \right| = 2$$

Question: If $\left| \frac{z}{z+i} \right| = 2$ represents a circle with centre P then distance of P from D is (where D:(1, 5))

Solution :

Let $z = x + iy$

$$\left| \frac{x+iy}{x+i(y+1)} \right| = 2$$

$$\Rightarrow \sqrt{x^2 + y^2} = 2\sqrt{x^2 + (y+1)^2}$$

$$x^2 + y^2 = [x^2 + y^2 + 2y + 1]$$

$$\Rightarrow 3x^2 + 3y^2 + 8y + 4 = 0$$

$$\Rightarrow x^2 + y^2 + \frac{8}{3}y + \frac{4}{3} = 0$$

Center $(0, \frac{-4}{3})$

Now, $P(0, \frac{-4}{3})$ and $D(1, 5)$

$$PD = \sqrt{1 + \left(\frac{-4}{3} - 5\right)^2} = \sqrt{1 + \frac{361}{9}}$$

$$= \sqrt{\frac{370}{9}}$$